

Agglomerative Methods Matrix Theory

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Matrix Updating Algorithm Scheme (MUAS)

- Initialization:

- $R_0 = \{\{x_i\}, i = 1, \dots, N\}$
- $P_0 = P(X)$
- $t = 0$

- Repeat:

- $t = t + 1$
- Find C_i, C_j such that

$$d(C_i, C_j) = \min_{r,s=1,\dots,N, r \neq s} d(C_r, C_s)$$

- Merge C_i, C_j into a single cluster C_q and form $R_t = (R_{t-1} - \{C_i, C_j\}) \cup \{C_q\}$
- Define the proximity matrix P_t from P_{t-1}

- Until R_{N-1} clustering is formed, that is, all vectors lie in the same cluster.

Update Algorithms

- The **single link algorithm**:

$$d(C_q, C_s) = \min\{d(C_i, C_s), d(C_j, C_s)\}$$

The single link has a tendency to favor **elongated clusters**.

- The **complete link algorithm**:

$$d(C_q, C_s) = \max\{d(C_i, C_s), d(C_j, C_s)\}$$

The complete link algorithm proceeds by recovering small **compact clusters**.

- The **Ward or minimum variance** algorithm. Here, the distance between two clusters C_i and C_j , d_{ij} , is defined as a weighted version of the squared Euclidean distance of their mean vectors, that is,

$$d_{ij} = \frac{n_i n_j}{n_i + n_j} d_{ij}^2$$

where $d_{ij}^2 = \|m_i - m_j\|^2$.